Using Maimonides' Rule to Estimate the Effect of Class Size on Scholastic Achievement

Angrist & Lavy (1999), *QJE* 114(2)

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ECX5479 Empirical Strategies for Policy Evaluation

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Roadmap

Part	Content	Presenter
I	Setup: the question, the rule, the data	Otis Sandford
II	Estimation: from OLS to IV	Ruijie Chen
III	Credibility and meaning	Tianlin Yan

Part I

Setup

Otis Sandford

Motivation

- Do smaller classes raise test scores?
- Central to school-resource policy
- Evidence is mixed
  - Observational studies: weak effects (Hanushek)
  - Tennessee STAR experiment: positive effects
- This paper: a natural experiment from an administrative rule

Institutional background

- Israel, since 1969: maximum class size of 40
- Pupils attend their neighbourhood school
- Private elementary schooling is rare
- June 1991: national tests for all 4th and 5th graders

Maimonides' rule

$$f_{sc} = \frac{e_s}{\text{int}\left(\frac{e_s - 1}{40}\right) + 1}$$

Enrollment e_s	Classes	Predicted size f_{sc}
40	1	40
41	2	20.5
80	2	40
81	3	27

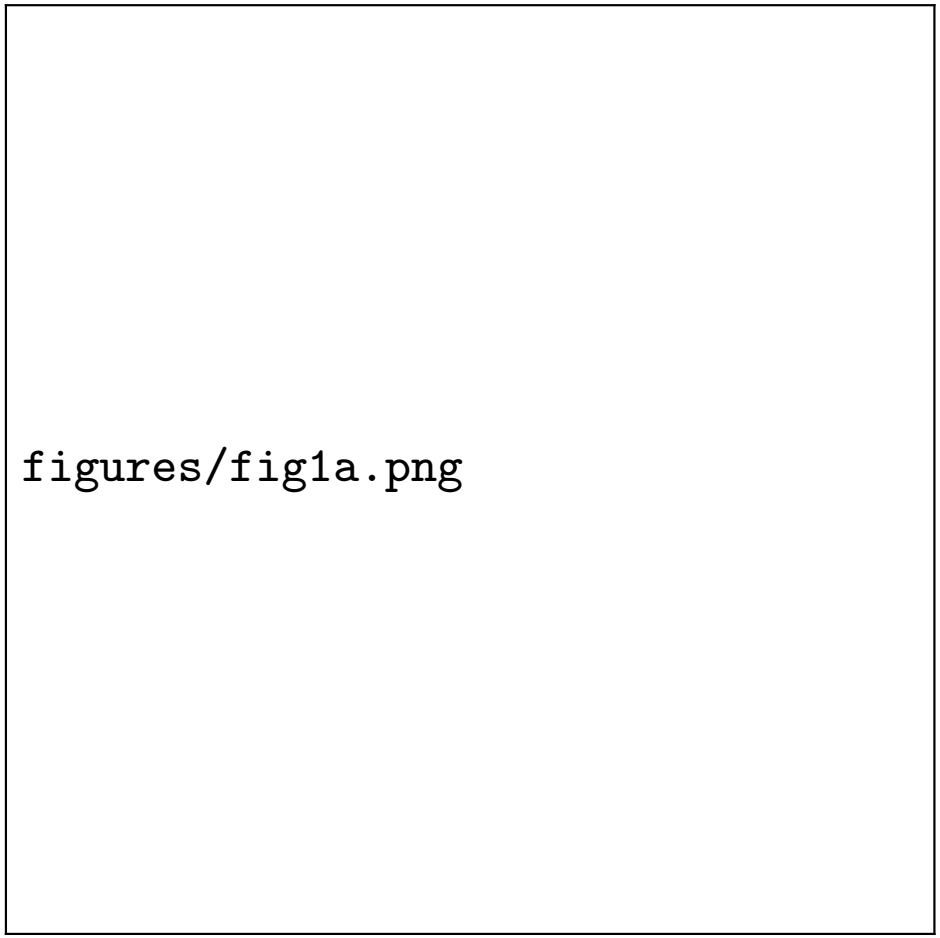
Class size jumps **down** as enrollment crosses 40, 80, 120

Data and sample

	Full sample		±5 discontinuity	
	5th	4th	5th	4th
Class size	29.9	30.3	30.8	31.1
Enrollment	77.7	78.3	76.4	78.5
Percent disadvantaged	14.1	13.8	13.6	12.9
Average verbal	74.4	72.5	74.5	72.5
Average math	67.3	68.9	67.0	68.7
Classes	2019	2049	471	415
Schools	1002	1013	224	195

Unit = class. Jewish public schools. Source: Table I (means).

The rule predicts actual class size



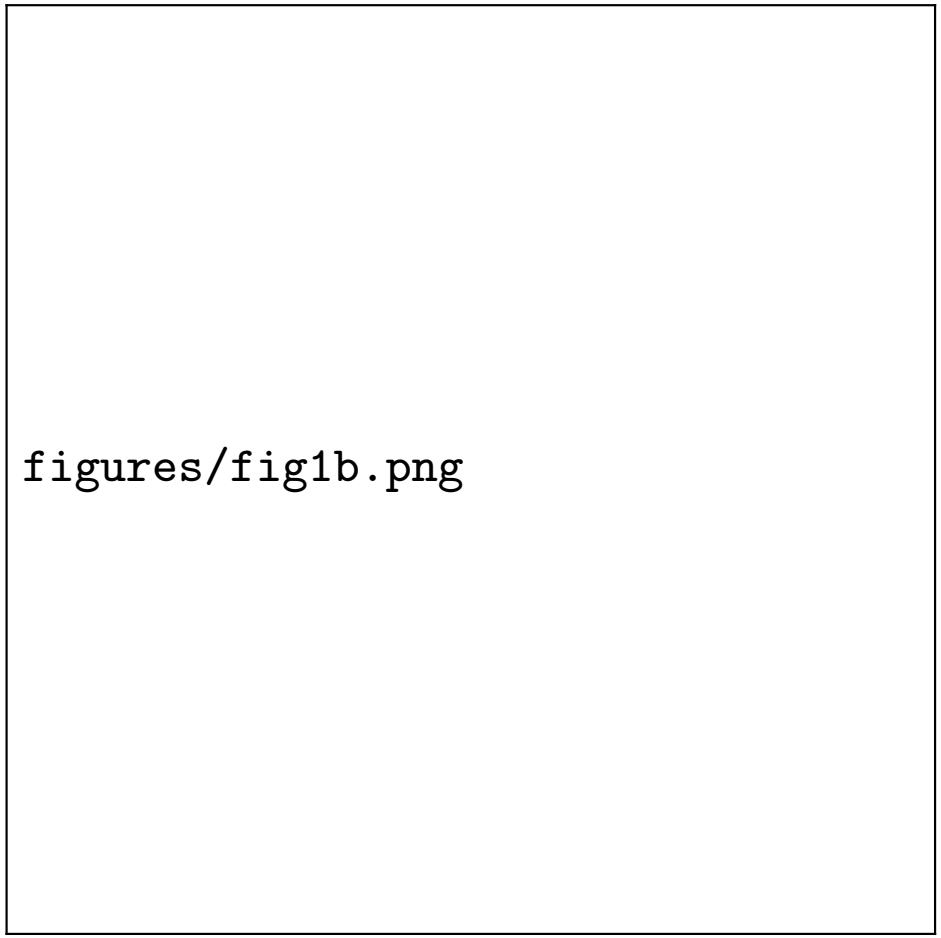
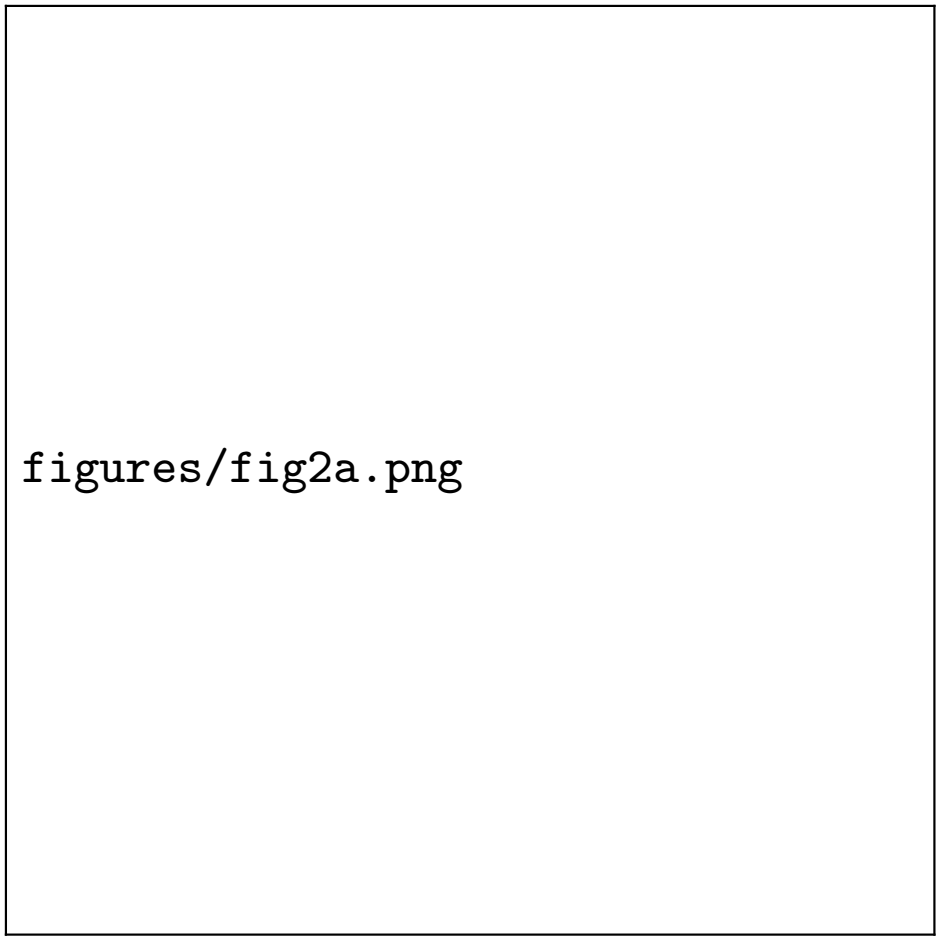


Figure I. Dashed = Maimonides' rule; solid = actual average class size.

Test scores mirror the rule



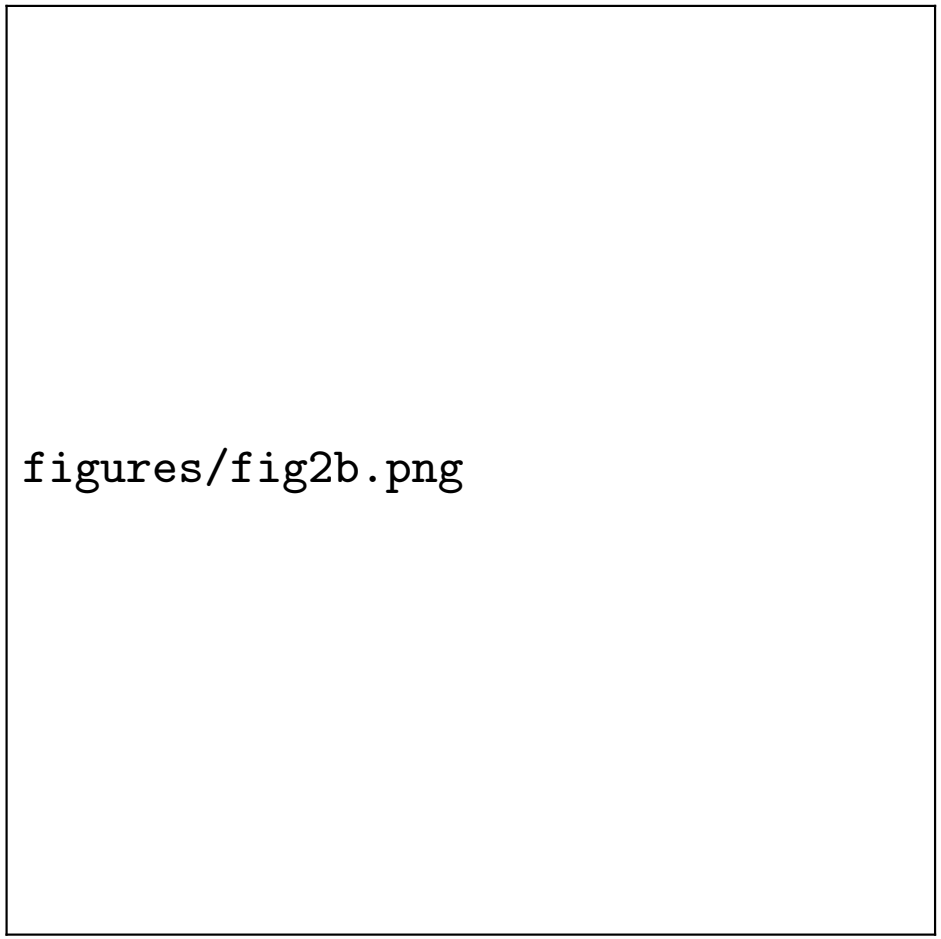


Figure II. Average reading score and predicted class size, enrollment bins of 10.

Suggestive. But is it causal?

Part II

Estimation

Ruijie Chen

Endogeneity: OLS is confounded

- Class size is endogenous: driven by family background and school size
- **Percent disadvantaged**: poorer schools get extra classes and score lower
- **Enrollment**: larger schools are in richer cities and have larger classes
- No controls: larger classes, *higher* scores. $\Rightarrow$ Counterintuitive
- With controls: coefficients shrink toward zero $\Rightarrow$ raw OLS is biased

	5th grade reading			5th grade math		
	(1)	(2)	(3)	(4)	(5)	(6)
Class size	**.221**	−.031	−.025	**.322**	**.076**	.019
	(.031)	(.026)	(.031)	(.039)	(.036)	(.044)
% disadvantaged		✓	✓		✓	✓
Enrollment			✓			✓

4th grade: same pattern. Bold: significant at 5%. Source: Table II.

Identification: fuzzy RD as IV

- **Why fuzzy RD**: the rule shifts class size at 40, 80, 120, but does not fix it
- **Predicted class size f_{sc} as IV**
  - **Relevance**: f_{sc} moves actual class size (Table III)
  - **Exclusion**: given PercentageDisadvantaged and Enrollment, f_{sc} affects scores only via class size; In RD terms: other determinants of scores are **continuous** at the cutoffs (untestable)
- **First stage**

$$n_{sc} = X_s'\pi_0 + \pi_1 f_{sc} + \xi_{sc}$$

  Reduced form

$$\bar{y}_{sc} = X_s'\gamma + \rho\, f_{sc} + \eta_s + u_{sc}$$

- Control for PD and functions of enrollment X_s
- Effect of **rule-induced** class size: $\hat{\alpha} = \dfrac{\text{reduced form}}{\text{first stage}} = \dfrac{\hat{\rho}}{\hat{\pi}_1}$

First stage and reduced form

	5th grade		4th grade	
	Full	±5	Full	±5
Class size	**.542**	**.346**	**.670**	**.503**
	(.027)	(.052)	(.025)	(.053)
Reading	**−.149**	**−.202**	**−.089**	−.075
	(.035)	(.054)	(.040)	(.063)
Math	**−.124**	**−.154**	−.033	.012
	(.049)	(.077)	(.047)	(.080)

Controls: % disadvantaged and linear enrollment. Source: Table III.

- Strong first stage ($t \approx 20$): no weak-instrument problem
- 4th grade loses significance in the ±5 window ⇒ stricter test in Part III

2SLS: fifth graders

	Full sample				±5 sample	
	(1)	(2)	(3)	(4)	(5)	(6)
Reading	**−.158**	**−.275**	**−.260**	−.186	**−.410**	**−.582**
	(.040)	(.066)	(.081)	(.104)	(.113)	(.181)
Math	−.013	**−.230**	**−.261**	−.202	−.185	−.443
	(.056)	(.092)	(.113)	(.131)	(.151)	(.236)
% disadvantaged	✓	✓	✓		✓	✓
Enrollment		linear	quad.			linear
Piecewise trend				✓		

Source: Table IV.

- Baseline (2): significantly negative. 10 fewer pupils ⇒ +2.75 reading points
- More flexible enrollment controls (3)–(4): same sign

Fourth vs fifth grade

	5th grade	4th grade
Reading	✓	✓
Math	✓	—
Effect of -8 pupils (reading)	$0.29\,\sigma$	$0.13\,\sigma$

σ = SD of class-average scores. Source: Section V.B.

- Reading: effect in both grades; math: 5th grade only
- 5th grade effect is about twice as large (0.29σ vs 0.13σ, 8-pupil cut)
- Possible reason: cohorts move through school together, so effects may accumulate

Part III

Credibility and meaning

Tianlin Yan

A closer comparison around the cutoffs

- Narrow windows reduce reliance on the functional form of enrollment controls.
- ± 5 sample: $[36, 45]$, $[76, 85]$, $[116, 125]$.
- ± 3 sample: $[38, 43]$, $[78, 83]$, $[118, 123]$.

Three segment-specific instruments

$$z_{jsc} = 1[c_j + 1 \leq e_s \leq c_j + h], \qquad c_j \in \{40, 80, 120\}, \quad h \in \{5, 3\}.$$

- Controls: two segment dummies; no continuous enrollment trend.
- Compare specifications with and without PD.

Intuition within a single segment, without other covariates

$$\widehat{\alpha}_{\text{Wald}} = \frac{\overline{y}_{\text{above}} - \overline{y}_{\text{below}}}{\overline{n}_{\text{above}} - \overline{n}_{\text{below}}}$$

Pooled IV combines evidence across the three segments.
Source: Angrist & Lavy (1999), pp. 557–559.

Table VI: strongest support for fifth-grade reading

	Reading			Math		
Window	±5	±3	±3	±5	±3	±3
Fifth grade	−0.687*	−0.588*	−0.451	−0.596*	−0.395	−0.270
	(0.197)	(0.198)	(0.236)	(0.254)	(0.254)	(0.281)
Fourth grade	−0.175	−0.234	−0.380	0.018	−0.118	−0.247
	(0.130)	(0.157)	(0.205)	(0.162)	(0.202)	(0.234)
PD control	Yes	Yes	No	Yes	Yes	No
Segment controls	Yes	Yes	Yes	Yes	Yes	Yes
N: fifth grade	471	302	302	471	302	302
N: fourth grade	415	265	265	415	265	265

Class-size coefficients; standard errors corrected for within-school correlation in parentheses.
* Significant at 5% using $|\widehat{\alpha}/\mathrm{SE}| > 1.96$.

- **Fifth grade:** reading remains significant with PD; math is less precise in narrower windows.
- **Fourth grade:** no estimate is significant at 5% here.

Source: Selected rows from Angrist & Lavy (1999), Table VI, p. 558.

Internal validity: what could undermine identification?

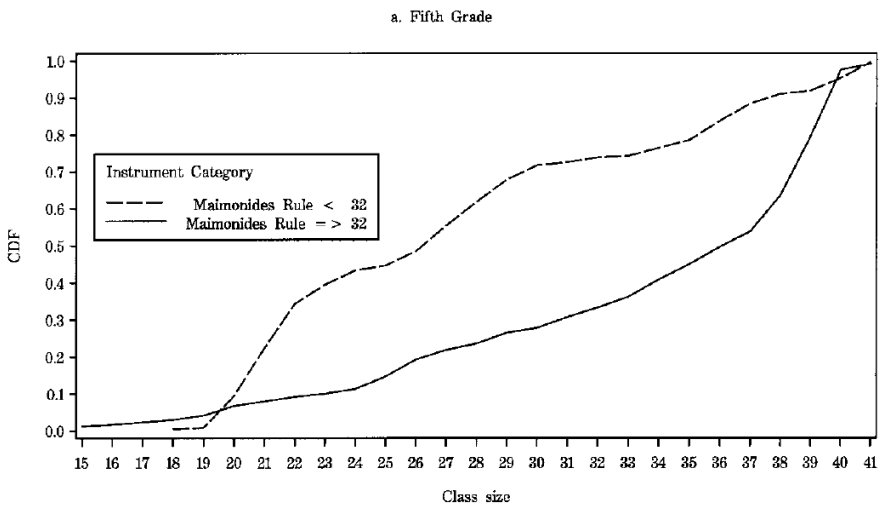
- **Key assumption:** pupil characteristics and other determinants of achievement vary smoothly across enrollment cutoffs.
- **Why plausible:** enrollment is difficult to predict and school choice is constrained by residence. September enrollment also limits concerns about later withdrawals.
- **Threat 1 — selective sorting:** if families or schools selectively adjust enrollment around cutoffs, pupil composition could jump alongside class size.
- **Threat 2 — other changes at the cutoff:** if opening another class also changes teacher quality or pupil grouping, the score difference may reflect more than class size alone.

Our assessment

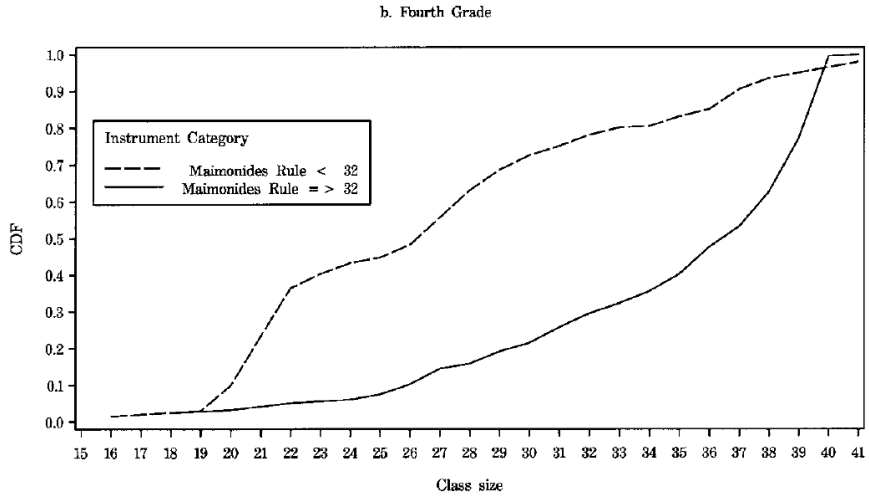
Narrow-window results strengthen credibility, but cannot rule out sorting or other simultaneous changes at the cutoffs.

Source: Angrist & Lavy (1999), pp. 549–550, 557–559.
Potential threats are our assessment, not established findings.

External validity: which class-size changes matter?



Fifth grade



Fourth grade

- Largest CDF gaps: 22–36 pupils, especially 28–35.
- Fifth-grade first-stage class-size differences: 10.7, 4.4 and 1.1 pupils near the 40, 80 and 120 cutoffs, respectively (controlling for PD and segment effects).
- Larger first-stage shifts give smaller schools more weight in the IV estimates.

Figure IV uses $Z = 1 - z_{sc}$: $Z = 1$ predicts larger classes.
Source: Angrist & Lavy (1999), pp. 565–566.

Policy implications and limits

The paper's policy illustration

- Reduce the class-size cap from 40 to 30.
- Average size: about 31 $\rightarrow$ 25.
- About 600 additional classes per grade.
- Potential gains: roughly two deciles in the distribution of class averages.
- Benefits are supported most clearly for fifth graders; fourth-grade evidence is weaker.

Why extrapolation is uncertain

- Main evidence: 1991 fourth and fifth graders; class-average scores.
- Third-grade results in 1992 are inconclusive; testing conditions changed.
- Israeli classes were relatively large by US standards at the time.

A policy projection, not an evaluated reform or a cost-benefit analysis.

Source: Angrist & Lavy (1999), pp. 562–563, 567, 569–570.

Conclusion and policy

- **What we learn:** smaller classes improve achievement most clearly for fifth graders; fourth-grade evidence is weaker.
- **Where the evidence applies:** rule-induced class-size changes in Israeli public schools, with more weight on smaller schools and class-size margins around 22–36 pupils.
- **How I would implement it:** introduce a phased pilot in schools with similar class sizes, while monitoring teacher recruitment and classroom capacity.
- **What I would evaluate before expansion:** compare learning gains and costs with a credible comparison group, and assess whether benefits extend across grades.

My policy judgment

I would support a targeted, evaluated class-size reduction. These local estimates alone do not justify a universal cap of 30.

Evidence: Angrist & Lavy (1999), Sections V–VI.
The phased pilot is my policy recommendation.

Thank you